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QUALITY INSPECTION APPARATUS AND INSPECTION SYSTEM

Abstract

A quality inspection apparatus capable of communicating with at least an image forming apparatus includes a first reception unit that receives from the image forming apparatus a first image generated by execution of image generation processing on print data by the image forming apparatus from the image forming apparatus, a first inspection unit that performs first inspection based on a scanned image acquired by reading of a printed product printed by the image forming apparatus and the first image, a second inspection unit that performs second inspection based on a second image generated by execution of image generation processing on the print data by an image processing apparatus and the first image, and a transmission unit that transmits information based on a result of inspection performed by the first inspection unit and a result of inspection performed by the second inspection unit to the image forming apparatus.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of International Patent Application No. PCT/JP2023/042137, filed Nov. 24, 2023, which claims the benefit of Japanese Patent Application No. 2022-188904, filed Nov. 28, 2022, both of which are hereby incorporated by reference herein in their entirety.

BACKGROUND OF THE INVENTION
Field of the Invention
[0002] The present invention relates to a quality inspection apparatus and an inspection system.
Background Art
[0003] A digital print technique based on electrophotography or the like, which is called print-on-demand, has become common mainly in the print industry, and there is a need for maintaining print quality and effectively producing printed products. To maintain print quality, a quality inspection apparatus that inspects image quality of printed products has been devised in late years.
[0004] According to Japanese Patent Application Laid-Open No. 2015-53561, when making determination, a quality inspection apparatus first registers image data generated by a raster image processor (RIP) in a print apparatus as a print sample image. The quality inspection apparatus uses image data, which is obtained by electronic reading of a medium on which image data has been printed, as an inspection target image, to determine whether the print sample image and the inspection target image are matched with each other, and checks whether there is no defect in a printed product.
[0005] A quality inspection apparatus that inspects image quality of a printed product performs image quality inspection based on image data generated by RIP processing that is appropriate for a print apparatus. Meanwhile, in industrial print, checking or the like of a document that is desired to be printed is performed based on a printed image subjected to RIP processing by a print apparatus that is used by a user in normal business operations or a RIP apparatus. The RIP apparatus has characteristics depending on each program included in the RIP apparatus and there may be a slight difference in color, image position, or the like.
CITATION LIST
Patent Literature
[0006] PTL 1: Japanese Patent Application Laid-Open No. 2015-53561
SUMMARY OF THE INVENTION
[0007] According to an aspect of the present invention, a quality inspection apparatus capable of communicating with at least an image forming apparatus includes a reception interface (IF) configured to receive from the image forming apparatus a first image generated by execution of image generation processing on print data by the image forming apparatus, a controller having a processor which executes instructions stored in a memory or having circuitry, the controller being configured to execute first inspection based on a scanned image acquired by reading of a printed product printed by the image forming apparatus and the first image, and execute second inspection based on a second image generated by execution of image generation processing on the print data by an image processing apparatus and the first image, and a transmission IF configured to transmit

information based on an inspection result of the first inspection and an inspection result of the second inspection to the image forming apparatus.
[0008] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] FIG. **1** is an example of a schematic diagram illustrating an overall configuration for describing an exemplary embodiment.
[0010] FIG. **2** is an example of a diagram illustrating a print apparatus and a quality inspection apparatus for describing the exemplary embodiment.
[0011] FIG. **3** is an example of a schematic diagram illustrating an overall hardware configuration for describing the exemplary embodiment.
[0012] FIG. **4**A is an example of a diagram illustrating each program for describing the exemplary embodiment.
[0013] FIG. **4**B is an example of a diagram illustrating each program for describing the exemplary embodiment.
[0014] FIG. **4**C is an example of a diagram illustrating each program for describing the exemplary embodiment.
[0015] FIG. **4**D is an example of a diagram illustrating each program for describing the exemplary embodiment.
[0016] FIG. **4**E is an example of a diagram illustrating each program for describing the exemplary embodiment.
[0017] FIG. **5** is an example of a sequence chart for describing registration of a print sample image according to a first exemplary embodiment.
[0018] FIG. **6** is an example of a sequence chart for describing inspection according to the first exemplary embodiment.
[0019] FIG. **7** illustrates an example of a screen for registering a print sample image according to the first exemplary embodiment.
[0020] FIG. **8**A illustrates an example of a screen for making inspection settings according to the first exemplary embodiment.
[0021] FIG. **8**B illustrates an example of a screen for making inspection settings according to the first exemplary embodiment.
[0022] FIG. **9**A illustrates an example of a screen for displaying an inspection result according to the first exemplary embodiment.
[0023] FIG. **9**B illustrates an example of a screen for displaying an inspection result according to the first exemplary embodiment.
[0024] FIG. **9**C illustrates an example of a screen for displaying an inspection result according to the first exemplary embodiment.
[0025] FIG. **10** is a flowchart for describing step S**2007**.
[0026] FIG. **11** is a flowchart for describing step S**3002**.
[0027] FIG. **12** is a flowchart for describing step S**3003**.
[0028] FIG. **13** is a flowchart for describing step S**2010**.
[0029] FIG. **14** is an example of a sequence chart for describing a second exemplary embodiment.
[0030] FIG. **15** is an example of a flowchart for describing a third exemplary embodiment.
DESCRIPTION OF THE EMBODIMENTS
[0031] A print quality inspection apparatus performs print inspection by comparing a sample image created by raster image processor (RIP) processing in a print apparatus and a printed image

subjected to RIP processing in the print apparatus, and is thereby capable of inspecting scratches, smudges, and the like on a printed product. Meanwhile, the print quality inspection apparatus is not capable of performing inspection in reflection of a difference in characteristics between RIP processing performed in the print apparatus and RIP processing performed in a print apparatus that is used by a user in normal business operations or a RIP apparatus. Thus, the print quality inspection apparatus has determined that a result is OK even if a deliverable is different in color or shifted in an image position from a printed product as expected by the user, that is, a printed product subjected to RIP processing.

[0032] One object of the present invention is to perform print inspection on a printed material as expected by the user. Another object of the present invention is to output an inspection result in consideration of a difference between image data generated by RIP processing in the RIP apparatus and image data generated by RIP processing in the print apparatus.

[0033] A best mode for implementing the present invention will be described below with reference to drawings.

[0034] FIG. **1** is a diagram illustrating an overall configuration of an image inspection system (printed product inspection system) for describing an exemplary embodiment. In the following description, a print apparatus may also be referred to as an image forming apparatus, a multi-function apparatus or a multi-function peripheral (MFP). Additionally, an external RIP apparatus according to the present exemplary embodiment represents an apparatus that performs RIP processing that is different from RIP processing performed by a print apparatus **100**, and may be also referred to as an image processing apparatus or a RIP apparatus. Although it is called the external RIP apparatus, the external RIP apparatus is not necessarily an apparatus that performs only RIP processing and may be a print apparatus that performs RIP processing and then executes print.

[0035] A personal computer (PC) **300** and a PC **600** are connected to the print apparatus **100** by a local area network (LAN) **400**. Furthermore, a quality inspection apparatus **200** is connected to the LAN **400**. The print apparatus **100** and the quality inspection apparatus **200** are connected to each other so that a medium can be directly conveyed therebetween. The print apparatus **100** and the quality inspection apparatus **200** may also be collectively referred to as a print quality inspection apparatus. Furthermore, an external RIP apparatus **500** is connected to the LAN **400**.

[0036] The external RIP apparatus **500** may be a RIP apparatus that is used by a user who uses the image inspection system at the time of offset print or a RIP apparatus that is used for proof print. However, the external RIP apparatus **500** is the one that is capable of converting print data such as page description language (PDL) data or portable document format (PDF) data into image data as expected by the user.

[0037] FIG. **2** is a diagram illustrating a detailed configuration regarding the print apparatus **100** and the quality inspection apparatus **200**. Image forming stations **101** to **104** are used to perform color print in yellow, magenta, cyan, and black, respectively. As the image forming stations **101** to **104**, it is conceivable to employ an image forming means using electrophotography, an inkjet method, or the like, but the present invention is not limited thereto. Additionally, the present invention can be implemented by a configuration of performing monochrome print, for example, a configuration including only the image forming station **104** for black.

[0038] A paper feeding device **114** is configured to include a paper feeding deck, and is capable of feeding a recording sheet such as paper from a paper feeding unit **105** or **106** to perform print on the recording sheet.

[0039] An intermediate transfer belt **108** rotates, and color materials are transferred to the intermediate transfer belt **108** from the image forming stations **101** to **104**. Furthermore, the color materials are transferred in a contact portion with a medium conveyed on a paper conveying path **109** toward the downstream side.

[0040] The paper conveying path **109** is connected to the print apparatus **100** and the quality inspection apparatus **200**, and the printed medium is directly conveyed to the inside of the quality inspection apparatus **200**.

[0041] A pair of image sensors **110** is installed with the paper conveying path **109** interposed therebetween, and captures images of both sides of the medium on the paper conveying path **109**. The images captured by the image sensors **110** are used for registration of a print sample image or execution of inspection. A plurality of paper discharge units is installed, and a printed product is sorted into paper discharge units **112** and **113** based on an inspection result. For example, a printed product whose inspection result is determined as OK is discharged to the paper discharge unit **112**, and a printed product whose inspection result is determined as a fail is discharged to the paper discharge unit **113** for purging.

[0042] In the present exemplary embodiment, the description is being given of the case where the quality inspection apparatus **200** controls a paper discharge destination of a printed product based on an inspection result, but the configuration is not limited thereto. For example, the image inspection system may employ a configuration in which a quality inspection apparatus makes notification about an inspection result and a print apparatus including a paper discharge unit controls a paper discharge destination of a printed product based on the inspection result.

[0043] FIG. **3** illustrates a circuit configuration of each apparatus.

[0044] The print apparatus **100** is connected to the LAN **400** via a network controller **120**. Additionally, processing executed by the print apparatus **100** is implemented by a central processing unit (CPU) **122** loading a program, which is stored in a storage device **121**, in a memory **123** and executing the program. The print apparatus **100** includes an operation unit **124** and is capable of performing display on a screen to accept an input operation from the outside. The print apparatus **100** further includes an image processing unit **125**. The image processing unit **125** converts electronic image data (for example, multi-value image data of Commission Internationale de L'Eclairage (CIE)-standard red, green, and blue (sRGB) into electronic image data for print (for example, a cyan (C), magenta (M), yellow (Y), and black (K) halftone image). Furthermore, the electronic image data for print is transferred to a print processing unit **126**, and the print apparatus **100** uses the image forming stations **101** to **104** to transfer the electronic image data for print to recording paper fed from the paper feeding device **114** and performs print.

[0045] The quality inspection apparatus **200** is connected to the LAN **400** via a network controller **221**. Additionally, processing executed by the quality inspection apparatus **200** is implemented by a CPU **223** loading a program, which is stored in a storage device **222**, in a memory **224** and executing the program. Furthermore, the quality inspection apparatus **200** uses a reading unit **225** connected to the image sensors **110** to convert an image on the recording paper conveyed through the paper conveying path **109** into electronic image data (for example, R, G, and B multi-value image data). An operation unit **226** functions as a display unit that displays a print sample, which will be described below, and an inspection setting screen, and also accepts input from a user. Screens displayed on the display unit are controlled by the CPU **223**, and the CPU **223** may be referred to as a display control unit in the present exemplary embodiment.

[0046] The PC **300** for giving an instruction for print to the print apparatus **100** is connected to the LAN **400** via a network controller **301**. Additionally, processing executed by the PC **300** is implemented by a CPU **303** loading a program, which is stored in a storage device **302**, in a memory **304** and executing the program. Furthermore, an operation unit **305** is connected to a display (not illustrated), and allows for screen display. Furthermore, the operation unit **305** is connected to a mouse and a keyboard (not illustrated), and allows for an operation of a program.

[0047] The PC **600** to be used solely for RIP inspection is connected to the LAN **400** via a network controller **601**. Additionally, processing executed by the PC **600** is implemented by a CPU **603** loading a program, which is stored in a storage device **602**, in a memory **604** and executing the program. Furthermore, an operation unit **605** is connected to a display (not illustrated), and allows for screen display. Furthermore, the operation unit **605** is connected to a mouse and a keyboard (not illustrated), and allows for an operation of a program. The PC **600** will be used in description of a second exemplary embodiment.

[0048] The external RIP apparatus **500** is connected to the LAN **400** via a network controller **501**. Additionally, processing executed by the external RIP apparatus **500** is implemented by a CPU **503** loading a program, which is stored in a storage device **502**, in a memory **504** and executing the program.

[0049] FIGS. **4**A to **4**D illustrate programs stored in the storage devices **121**, **222**, **302**, **502**, and **602**, respectively. FIG. **4**A illustrates programs stored in the print apparatus **100**. Similarly, FIG. **4**B illustrates programs stored in the quality inspection apparatus **200**, FIG. **4**C illustrates programs stored in the PC **300**, FIG. **4**D illustrates a program stored in the external RIP apparatus **500**, and FIG. **4**E illustrates a program stored in the PC **600**. Details of the programs illustrated in FIGS. **4**A to **4**E will be described below.

[0050] While the quality inspection apparatus **200** is provided with the operation unit **226**, it is assumed that the CPU **223** has a function of generating a Hypertext Markup Language (HTML) for a screen. Thus, it is possible to perform display on the operation unit **305** in the PC **300** and operate the operation unit **305** using a Hypertext Transfer Protocol (HTTP). Hence, in the present invention, an apparatus that performs display and accepts an operation is not specifically limited.

First Exemplary Embodiment

[0051] A first exemplary embodiment is now described with reference to a sequence chart in FIG. **5**. In the print apparatus **100**, the quality inspection apparatus **200**, the PC **300**, and the external RIP apparatus **500**, processing is implemented by loading of each program, which is stored in a corresponding one of the storage devices **121**, **222**, **302**, and **502**, in a corresponding RAM and execution of the program by a corresponding one of the CPUs **122**, **223**, **303**, and **503**. Additionally, when transmitting or receiving data such as an image or a command, the print apparatus **100**, the quality inspection apparatus **200**, the PC **300**, and the external RIP apparatus **500** respectively use the network controllers **120**, **221**, **301**, and **501** to access the data or the command and store and read out the data or the command through the LAN **400**. A description about a series of operations is omitted.

<Registration of Print Sample>

[0052] First, a method of registering a print sample for inspection is described with reference to the sequence chart in FIG. **5**.

[0053] In step S**1001**, when accepting an instruction for print from the operation unit **305**, the CPU **303** in the PC **300** executes a print data creation program **310** and creates print data. The print data created here is composed of an image such as a Joint Photographic Experts Group (JPEG) image or a Tag Image File Format (TIFF) image, text data, font information to be used for the text data, graphics drawing data, and the like. Furthermore, the CPU **303** executes a print setting program **311**, acquires print settings included in the accepted print instruction, and sets the print settings in the print data. The print settings include, for example, settings unique to the print apparatus **100** such as settings regarding from which paper feeding unit (the paper feeding unit **105** or **106**) in the paper feeding device **114** paper is fed.

[0054] Subsequently, in step S**1002**, the CPU **303** in the PC **300** transmits the print data created in step S**1001** to the print apparatus **100** via the LAN **400**.

[0055] In step S**1003**, the CPU **122** in the print apparatus **100** executes a RIP processing program **130**, performs RIP processing on received print data, and creates image data. The RIP processing mentioned herein is raster image processor processing, and represents processing of generating image data from received print data. Specifically, the RIP processing is to interpret a page description language (PDL), which is used in text data and image data, and convert the text data and the image data into a raster image. In the present exemplary embodiment, the RIP processing is referred to as image generation processing. The image data created here is electronic image data composed of color information such as C, M, Y, and K. In a case of print data in a plurality of

pages, electronic image data for all the pages is created.
[0056] In step S**1004**, the CPU **122** in the print apparatus **100** transmits the image data created in step S**1003** to the quality inspection apparatus **200** via the LAN **400**.
[0057] In step S**1005**, the CPU **223** in the quality inspection apparatus **200** executes sample image creation processing **236**. Specifically, the CPU **223** creates a print sample image for image inspection based on the image data received in step S**1004** and displays the print sample image on a screen **700**. For example, in a case where the image data is a CMYK image, the CPU **223** performs color conversion processing using an International Color Consortium (ICC) profile preliminarily stored in the storage device **222** to convert the image data into RGB image data, and thereby creates the print sample image. FIG. **7** illustrates an example of the screen **700** for approving creation of the print sample image to be displayed on the operation unit **226**. When the CPU **223** detects pressing of an OK button **701** from the operation unit **226**, the processing proceeds to step S**1006**. In contrast, when detecting pressing of a cancel button **702** from the operation unit **226**, the CPU **223** stops all processing, and ends creation of the print sample image.
[0058] In step S**1006**, the CPU **223** in the quality inspection apparatus **200** executes inspection registration processing **237**. Specifically, the CPU **223** registers an inspection setting accepted from an inspection setting screen illustrated in each of FIGS. **8**A and **8**B. On the inspection setting screen, the CPU **223** accepts the inspection setting while displaying the print sample image created in step S**1005**. FIGS. **8**A and **8**B each illustrate the inspection setting screen as an example. FIG. **8**A illustrates an inspection setting screen for defect inspection. FIG. **8**B illustrates an inspection setting screen for RIP inspection.
[0059] The defect inspection in FIG. **8**A is to set a defect inspection level at the time of print. Examples of the defect at the time of print include, in a case of image formation in an electrophotographic process, toner smudges caused by insufficient cleaning in the image forming stations **101** to **104** or other factors. On the screen in FIG. **8**A, it is possible to designate a freely-selected range on a displayed print sample image, and freely set image inspection level settings **705** and **706** in this range.
[0060] For example, if each of the image inspection level settings **705** and **706** is set at level 2, the CPU **223** determines that a result is OK even if there is a certain amount of smudges. In contrast, if each of the image inspection level settings **705** and **706** is set at level 4, it means a strict setting in which the CPU **223** determines that a result is a fail even with slight smudges.
[0061] Meanwhile, the RIP inspection in FIG. **8**B represents a setting for inspecting whether the image data created by the RIP processing program **130** is correct image data. The RIP processing program is a program for interpreting the print data, and there are certain characteristics depending on a RIP processing program to be provided. For example, when text data is replaced by an image using a registered font and image data is created, there is a case where a font that is not included in a RIP processing program is replaced by a different font. Additionally, when a vector graphic pattern is drawn, a position of a drawn graphic pattern is different depending on characteristics of each RIP processing program. Furthermore, each RIP processing program has characteristics in interpretation of a layer structure, and there is a case where a color changes at the time of superimposition of drawn images. The screen in FIG. **8**B is used to set a permissible level for a difference due to characteristics of each RIP processing program to be used. On the screen in FIG. **8**B, it is possible to designate a freely-selected range on a displayed print sample image, and set an inspection level in a positional shift inspection setting **710** and an inspection level in a color inspection setting **711**. The inspection level is a permissible level. A plurality of displayed inspection ranges in FIG. **8**B allows for discrimination with a color of a displayed frame, a solid line, a ruled line, or the like. Additionally, the positional shift inspection and the color inspection can be performed in an identical region in an overlapping manner.
[0062] The screen illustrated in FIG. **8**A and the screen illustrated in **8**B can be switched by pressing of tabs in an upper stage.

[0063] When detecting pressing of an OK button **703** in FIG. **8**A or an OK button **708** in FIG. **8**B, the CPU **223** ends registration of the print sample image. At this time, the CPU **223** sets a name of inspection, and collectively stores the name of inspection together with the print sample image in the storage device **222** in the quality inspection apparatus **200** as a setting file.
<Inspection>
[0064] Subsequently, an inspection process is described with reference to FIG. **6**.
[0065] In step S**2001**, when detecting the start of inspection, the CPU **223** in the quality inspection apparatus **200** reads out the print sample image and the setting file, which have been created in steps S**1005** and S**1006**, from the storage device **222**.
[0066] In step S**2002**, when accepting an instruction for print from the outside, the CPU **303** in the PC **300** executes the print data creation program **310** and creates print data. The present processing may be similar to the processing in step S**1001**. Step S**2002** includes, in addition to the processing in step S**1001**, setting a number of copies, and allows for a print setting to print a plurality of copies.
[0067] In step S**2003**, the CPU **303** in the PC **300** transmits the print data created in step S**1001** to the print apparatus **100** and the external RIP apparatus **500** via the LAN **400**. In step S**2004**, the CPU **122** in the print apparatus **100** executes the RIP
[0068] processing program **130** and creates first image data. Specifically, the CPU **122** performs first RIP processing provided by the print apparatus **100** on the received print data, and creates first image data. In step S**2005**, the CPU **503** in the external RIP apparatus **500** executes a RIP processing program **510**, creates second image data, and stores the second image data in the storage device **502**. Specifically, the CPU **503** performs second RIP processing provided by the external RIP apparatus **500** on the received print data, and creates second image data.
[0069] In the present exemplary embodiment, the RIP processing program **130** and the RIP processing program **510** have an identical function, but are different programs. The RIP processing program **130** is designed solely for the print apparatus **100**, and allows for generation of image data in which resolution necessary for the print apparatus **100** or the like is optimally set. In the present exemplary embodiment, the RIP processing program is a program that is built into the print apparatus **100**, but may be a program built into another apparatus. Specifically, the image inspection system may have a separate RIP apparatus that performs RIP processing designed solely for the print apparatus **100**, and the print apparatus **100** may receive image data created by the other RIP apparatus.
[0070] A description is now given of a reason that the print apparatus **100** does not execute print with image data subjected to RIP processing with use of the RIP processing program **510** provided by the external RIP apparatus **500**. When creating image data from print data, the print apparatus **100** performs processing of making layout settings such as a media type and double-sided print. These layout settings are settings unique to the print apparatus **100**, and generally cannot be processed by the RIP processing program **510**. Hence, in a case where the print apparatus **100** executes print, it generally uses image data created with the RIP processing program **130**.
[0071] Meanwhile, the RIP processing program **510** allows for generation of image data serving as a sample for print. For example, the RIP processing program **510** is considered to be a RIP processing program that has been used to create image data used for an operation of proofreading a document before a print operation.
[0072] In step S**2006**, the CPU **503** in the external RIP apparatus **500** transmits the image data created in step S**2005** to the quality inspection apparatus **200** via the LAN **400**. Alternatively, in step S**2006**, the CPU **223** in the quality inspection apparatus **200** may read the second image data stored in the storage device **302** via the LAN **400**.
[0073] In step S**2007**, the CPU **223** in the quality inspection apparatus **200** executes a RIP inspection program **231**. The CPU **223** uses the RIP inspection program **231** to compare pages on which print processing of the first image data and the second image data is being executed and

check whether a result from the RIP processing program **130** is correct. The CPU **223** then outputs OK or a fail as a result of RIP processing. In a case where the result is a fail, the CPU **223** stores a defect image in the storage device **222**. Details of RIP inspection will be described below. In the present exemplary embodiment, the CPU **223** performs RIP inspection once before reading processing in step S**2009**, but a timing is not limited thereto. The CPU **223** may be configured to perform RIP inspection once or multiple times at a timing at which defect inspection is performed.

[0074] In step S**2008**, the CPU **122** in the print apparatus **100** executes print processing **131**. In the print processing **131**, the CPU **122** reads out a created RIP image in a storage region in step S**2004** and executes the print processing **131**. In the print processing **131**, after performing halftone processing on the RIP image in four colors (CMYK) and other processing, the CPU **122** places color materials on the intermediate transfer belt **108** in the image forming stations **101** to **104** and forms an image depending on the signal values. The formed image is transferred to recording paper, and conveyed immediately above/below the image sensors **110** for the front and back sides in the quality inspection apparatus **200** via the paper conveying path **109**.

[0075] In step S**2009**, the CPU **223** in the quality inspection apparatus **200** executes reading processing **232**, and reads the recording paper on which print has been performed in step S**2008** with the image sensors **110** for the front and back sides. The CPU **223** converts the read data into scanned image data, and stores the scanned image data as third image data in the storage device **222**.

[0076] In step S**2010**, the CPU **223** in the quality inspection apparatus **200** executes defect inspection **233** on the first and third image data. In the defect inspection, the CPU **223** compares a sample image and a scanned image to inspect whether there are print smudges on a printed product. Details of a defect inspection method will be described below with reference to FIG. **13**.

[0077] Since the third image data is data of an image obtained by reading of a medium printed by the print apparatus **100** with the image sensors **110**, it is possible to detect a defect in the print apparatus **100**. Details of the processing will be described below.

[0078] Image data created and used until step S**2010** is summarized in Table 1. It is preferable that resolution of each image data be identical, and the present exemplary embodiment is based on the assumption that resolution is 300 dpi.

TABLE-US-00001 TABLE 1 Apparatus that Program Image type creates image to be used First image Print apparatus Print RIP inspection data RIP image apparatus 100 231, defect inspection 233 Second image External RIP RIP apparatus RIP inspection data image 500 231 Third image Print product read Quality inspection Defect inspection data image apparatus 200 233

[0079] In step S**2011**, the CPU **223** in the quality inspection apparatus **200** executes collation processing **234** in response to the result in step S**2009**. In the collation processing, for example, the CPU **223** makes final determination according to the following Table 2, and controls an operation of each apparatus.

TABLE-US-00002 TABLE 2 Defect inspection = OK Defect inspection = Fail RIP inspection = Normal paper Purging discharge + OK discharge re-print RIP inspection = Purging discharge + Purging discharge + Fail stop stop

[0080] Normal paper discharge in Table 2 is processing of executing control of discharge of paper to the paper discharge unit **112** in the quality inspection apparatus **200**. Additionally, purging paper discharge is processing of executing control of discharge of paper to the paper discharge unit **113** in the quality inspection apparatus **200**. That is, the CPU **223** controls a paper discharge destination of a printed product as an inspection target depending on a result of inspecting the printed product, and can thereby sort printed products into a defective printed product that is determined as a fail in inspection and a printed product that is determined as OK in inspection. Combinations described in Table 2 may be preliminarily set as fixed values, or settings made by a system administrator may be accepted.

[0081] Even in a case where a result of the defect inspection is OK but if a result of the RIP inspection is a fail, the CPU **223** determines that the printed product is not appropriate as a deliverable, performs purging discharge, and stops print. The CPU **223** performs purging discharge because there is no smudge on the printed product, but the printed product is different in print position and/or color from a deliverable desired by the user. Furthermore, since there is an error in the RIP processing in the print apparatus **100**, the CPU **223** also stops print itself.

[0082] Additionally, in a case where a result of the defect inspection is a fail but if a result of the RIP inspection is OK, the CPU **223** performs purging discharge and re-print processing. The CPU **223** performs purging discharge because an image is appropriate as the deliverable desired by the user, but there are smudges on the printed product. However, since there is no error in the RIP processing itself, the CPU **223** performs re-print processing. The purging discharge mentioned herein represents discharge of the print product to a tray for a print product determined as a fail, which is different from a tray for a print product determined as OK, but is not limited thereto. In the purging discharge, the CPU **223** performs discharge so as to discriminate between a print product determined as OK in inspection and a print product determined as a fail in inspection.

[0083] In step S**2012**, the CPU **223** in the quality inspection apparatus **200** executes paper discharge processing **235** according to Table 2. In step S**2013**, after the completion of the processing in step S**2012**, the CPU **223** transmits the result to the print apparatus **100**. In a case of re-print according to Table 2, the CPU **223** re-executes processing in step S**2008** to perform re-print of an identical page. In this case, the CPU **223** in the quality inspection apparatus **200** re-executes processing steps S**2009** to S**2012**.

[0084] The CPU **223** repeatedly executes processing steps S**2004** to S**2013** until it completes print and inspection of all pages in the set number of copies. Additionally, in a case of determining to stop print in step S**2011**, the CPU **223** ends repetition processing. The processing then proceeds to step S**2014**.

[0085] In step S**2014**, the CPU **223** displays an inspection result. FIG. **9**A displays an example of display of a result. When detecting pressing of an OK button **720**, the CPU **223** ends inspection.

[0086] Additionally, in FIG. **9**A, buttons **721**, **722**, and **723** for displaying detected defect images are disposed. When detecting pressing of the button **721**, the CPU **223** displays a screen illustrated in FIG. **9**B. The screen illustrated in FIG. **9**B displays a defect at the time of print with use of the third image data. When detecting pressing of the button **723**, the CPU **223** displays a screen illustrated in FIG. **9**C. The screen illustrated in FIG. **9**C displays detection of a positional shift and/or detection of a fail regarding a color difference at the time of the RIP inspection with use of the first image data or the second image data.

[0087] In a case where there are both the defect at the time of print and detection of the positional shift and/or the fail regarding the color difference at the time of the RIP inspection, two buttons for displaying detected defect images may be prepared. In this case, prepared are a screen that displays the defect at the time of print with use of the third image data and a screen that displays detection of the positional shift and/or the fail regarding the color difference at the time of the RIP inspection with use of the first image data or the second image data. Additionally, the screen that displays the defect at the time of print with use of the third image data and the screen that displays detection of the positional shift and/or the fail regarding the color difference at the time of the RIP inspection with use of the first image data or the second image data may be displayed side by side on one screen. Alternatively, an image that merges the screen that displays the defect at the time of print with use of the third image data and the screen that displays detection of the positional shift and/or the fail regarding the color difference at the time of the RIP inspection with use of the first image data or the second image data may be created. In this case, the defect in print and the fail regarding the color difference and/or the positional shift may be superimposed on the merged image.

<RIP Inspection>

[0088] The RIP inspection program **231** executed by the CPU **223** in the quality inspection apparatus **200** in step S**2007** is now described with reference to FIG. **10**.

[0089] In step S**3001**, the CPU **223** first substitutes "inspection: OK" into a result as an initial value. In step S**3002**, the CPU **223** calculates a maximum value of a positional shift between the first and second image data. Regarding the positional shift, the CPU **223** calculates feature points in the first and second image data and compares respective positions. A method of calculating the maximum value of the positional shift will be described below.
[0090] In step S**3003**, the CPU **223** calculates a maximum value of a color difference between the first and second image data. A method of calculating the maximum value of the color difference will be described below.
[0091] In step S**3004**, the CPU **223** compares the maximum value of the positional shift calculated in step S**3002** and a threshold according to a positional shift inspection level set in step S**1006**. The threshold is stored in the storage device **222**, and is based on, for example, Table 3. Additionally, the positional shift inspection level is set by an administrator of the quality inspection apparatus **200**. Since the positional shift inspection level is set at level 2 in the screen example illustrated in FIG. **8**B, a positional shift up to 1 mm is permitted.
TABLE-US-00003 TABLE 3 Permissible level for positional shift Threshold for positional shift (mm) Level 1 2 Level 2 1 * Level 3 0.5 Level 4 0.1 * Default setting
[0092] As a result of comparison, if the positional shift is smaller than the threshold (YES in step S**3004**), the processing proceeds to step S**3007**. In contrast, if the positional shift is larger than the threshold (NO in step S**3004**), the processing proceeds to step S**3005**.
[0093] In step S**3005**, the CPU **223** stores, in the storage device **222**, an image created in step S**3002** and showing feature points indicating the positional shift. In step S**3006**, the CPU **223** substitutes "inspection: fail" into a result.
[0094] In step S**3007**, the CPU **223** compares the maximum value of the color difference calculated in step S**3003** and a threshold based on a value determined according to a color difference inspection level set in step S**1006**. The color difference is calculated by conversion into CIE-L*a*b*. The CPU **223** reads out an ICC profile for converting CMYK image data into an image in the CIE-L*a*b* space from the storage device **222** and performs color space conversion. The CPU **223** calculates the color difference by obtaining a Euclidean distance in the CIE-L*a*b* space, which will be described below. The threshold is stored in the storage device **222**, and is based on, for example, Table 4. Additionally, the color difference inspection level is set by the administrator of the quality inspection apparatus **200**. Since the color difference inspection level is set at level 2 in the screen example illustrated in FIG. **8**B, ΔE of up to 4 is permitted.
TABLE-US-00004 TABLE 4 Permissible level color difference Threshold for color difference (ΔE) Level 1 6 Level 2 4 * Level 3 3 Level 4 0 * Default setting
[0095] As a result of comparison, if the color difference is smaller than the threshold (YES in step S**3007**), the flow of the RIP inspection ends. In contrast, if the color difference is larger than the threshold (NO in step S**3007**), the processing proceeds to step S**3008**.
[0096] In step S**3008**, the CPU **223** stores, in the storage device **222**, an image created in step S**3003** and marking a region that is determined as a fail regarding the color difference. In step S**3009**, the CPU **223** substitutes "inspection: fail" into a result.
[0097] Subsequently, the method of calculating the maximum value of the positional shift in step S**3002** is now described with reference to a flowchart in FIG. **11**.
[0098] In step S**4001**, the CPU **223** reads out the first and second image data from the storage device **222**. In step S**4002**, the CPU **223** uses a feature quantity extraction method such as a scale invariant feature transform (SIFT) to extract feature points from the read first and second image data and acquires coordinates of the feature points.
[0099] In step S**4003**, the CPU **223** obtains coordinates of corresponding second feature points from the coordinates of the feature points extracted in step S**4002** from the first image data, and performs matching with use of a method such as Fast Library for Approximate Nearest Neighbors (FLANN). However, it is sufficient if the CPU **223** calculates only feature points at coordinates in a

positional shift inspection frame illustrated in FIG. **8**B.
[0100] In step S**4004**, the CPU **223** calculates a Euclidean distance between feature points matched in step S**4003** between the first and second image data. In step S**4005**, the CPU **223** obtains feature points whose Euclidean distance becomes a maximum distance among the feature points calculated in step S**4004**, and stores the feature points in the storage device **222**.
[0101] Subsequently, the method of calculating the maximum value of the color difference in step S**3003** is described with reference to a flowchart in FIG. **12**.
[0102] In step S**5001**, the CPU **223** reads out the first and second image data from the storage device **222**. In step S**5002**, the CPU **223** uses an algorithm such as a mean shift method with respect to the first and second image data to perform region division processing.
[0103] In step S**5003**, the CPU **223** performs matching of regions between the first and second image data. With this processing, determined is whether a division region created in step S**5002** in the first image data corresponds to a division region created in step S**5002** in the second image data. As a means of determination, the CPU **223** calculates a centroid of each region in the first and second image data to determine representative coordinates, and can determine regions close to the representative coordinates as corresponding regions.
[0104] In step S**5004**, the CPU **223** determines a representative color in the division region created in step S**5002**. The representative color is determined by calculation of an average of CMYK density values of pixels constituting each region and conversion into CIE-L*a*b*. In step S**5005**, the CPU **223** obtains a color difference in the CIE-L*a*b* space between corresponding regions obtained in step S**5003** in the first and second image data with respect to the representative color calculated in step S**5004** in each region, and stores a maximum color difference among obtained color differences in the storage device **222**.
[0105] The CPU **223** executes the RIP inspection in step S**2007** with use of the above-mentioned processing.
<Defect Inspection>
[0106] The defect inspection executed by the CPU **223** in step S**2010** is described with reference to FIG. **13**.
[0107] In step S**6001**, the CPU **223** reads out the first and third image data from the storage device **222**. In step S**6002**, the CPU **223** performs positional adjustment processing so that a position of the third image data is matched with a position of the first image data. As a method of positional adjustment processing, it is possible to perform positional adjustment by detecting a position of an end portion of paper for the third image data and rotating the third image data using an affine transformation method or the like so that the position of the end portion of paper for the third image data is matched with an assumed end portion of an image in the first image data.
[0108] In step S**6003**, the CPU **223** performs subtraction between the first and third images subjected to positional adjustment in step S**6002** and creates a difference image. Pixels remaining in this difference image can be determined as dusts or smudges that are supposed not to exist in an image.
[0109] In step S**6004**, the CPU **223** detects blobs from the difference image created in step S**6003**, and acquires a size of each blob. In the processing in step S**6004**, it is possible to make analysis using an image processing method called blob analysis.
[0110] In step S**6005**, the CPU **223** determines whether the largest blob among the blobs obtained in step S**6004** is smaller than the level determined in step S**1006**. If the largest blob is smaller than the level determined in step S**1006** (YES in step S**6005**), the processing proceeds to step S**6006**. In step S**6006**, the CPU **223** determines that a result of defect detection is OK. In contrast, if the largest blob is larger than the level determined in step S**1006** (NO in step S**6005**), the CPU **223** determines that a result of defect detection is a fail. The threshold is stored in the storage device **222**, and is based on, for example, Table 5. In Table 5, thresholds are defined based on the assumption of resolution of 300 dpi. Additionally, the inspection level is set by the administrator of

the quality inspection apparatus **200**.

TABLE-US-00005 TABLE 5 Inspection level Threshold for size of blob (pix) Level 1 60 Level 2 45 * Level 3 25 Level 4 15 * Default setting

[0111] As described above, the configuration of performing the RIP inspection and the defect inspection according to the first exemplary embodiment makes it possible to output a printed product without a defect as intended by the user and increase reliability of the print apparatus **100**.

[0112] In the present exemplary embodiment, the description has been given of the method in which one quality inspection apparatus **200** performs the RIP inspection and the defect inspection, but the number of quality inspection apparatuses **200** is not limited to one. For example, the image inspection system may employ a configuration in which two quality inspection apparatuses **200** are communicably connected to each other, the first quality inspection apparatus **200** performs the RIP inspection, and the second quality inspection apparatus **200** performs the defect inspection.

[0113] Additionally, in the present exemplary embodiment, the description has been given based on the assumption that the quality inspection apparatus **200** and the external RIP apparatus **500** are communicable via a network, but the quality inspection apparatus **200** and the external RIP apparatus **500** are not necessarily communicable. For example, the RIP processing in step S**2005** in FIG. **6** may be preliminarily performed in the external RIP apparatus **500**. Additionally, the external RIP apparatus **500** transmits image data to the quality inspection apparatus **200** in step S**2006**, but the image inspection system may have a configuration in which the quality inspection apparatus **200** receives image data subjected to RIP processing in the external RIP apparatus **500** using another method.

Second Exemplary Embodiment

[0114] A description is now given of a second exemplary embodiment in which the registration of the print sample image in the first exemplary embodiment is automatically performed, and, furthermore, the RIP inspection is performed by the PC **600** instead of the quality inspection apparatus **200** to improve operability of the quality inspection apparatus **200** and reduce processing load of the quality inspection apparatus **200**. A description of processing that is identical to that in the first exemplary embodiment is omitted as appropriate.

[0115] The second exemplary embodiment is described with reference to a sequence chart in FIG. **14**.

<Inspection>

[0116] In step S**7001**, when accepting an instruction for print from the outside or the operation unit **305**, the CPU **303** in the PC **300** executes the print data creation program **310** and creates print data. Since this operation is identical to the operation in step S**1001**, and a detailed description thereof is omitted.

[0117] In step S**7002**, the CPU **303** in the PC **300** transmits the print data created in step S**7001** to the print apparatus **100** and the external RIP apparatus **500** via the LAN **400**.

[0118] Since step S**7003** is identical to the processing in step S**2004** and step S**7004** is identical to the processing in step S**2005**, a detailed description thereof is omitted. Since step S**7005** is identical to the processing in step S**1004** and step S**7006** is identical to the processing in step S**2006**, a detailed description thereof is omitted.

[0119] In step S**7007**, the CPU **223** in the quality inspection apparatus **200** transfers two pieces of image data received in steps S**7005** and S**7006** to the PC **600**.

[0120] In step S**7008**, the CPU **603** in the PC **600** executes a RIP inspection program **610**. Since processing with the RIP inspection program **610** is identical to the processing with the RIP inspection program **231**, a detailed description thereof is omitted. However, in the second exemplary embodiment, the registration of the print sample image is not performed, and the settings in FIG. **8**B are not individually made. Inspection is performed on the whole image with use of default settings in Tables 3 and 4.

[0121] In step S**7009**, the CPU **603** in the PC **600** transfers a result in step S**7008** to the quality inspection apparatus **200**.

[0122] The image inspection system performs processing in steps S**7003** to S**7009** on all the pages. In a case where a determination result in step S**7009** is a fail, the image inspection system may stop the sequence in FIG. **14** and end processing.

[0123] In step S**7010**, the CPU **223** in the quality inspection apparatus **200** issues an instruction to start print to the print apparatus **100**.

[0124] Since processing in steps S**7011** to S**7012** is identical to the processing in steps S**2008** to S**2009**, a detailed description thereof is omitted.

[0125] Since processing in step S**7013** is identical to the processing in step S**2010**, a detailed description thereof is omitted. However, in the second exemplary embodiment, the registration of the print sample image is not performed, and the settings in FIG. **8**A are not individually made. The inspection is performed on the whole image with use of a default setting in Table 4.

[0126] Since processing in steps S**7014** to S**7017** is identical to the processing in steps S**2011** to S**2014**, a detailed description thereof is omitted.

[0127] As described above, the configuration of performing the RIP inspection and the defect inspection according to the second exemplary embodiment makes it possible to output a printed product without a defect as intended by the user and increase reliability of the print apparatus **100**.

Third Exemplary Embodiment

[0128] In a third exemplary embodiment, a description is given of a method of using image data created by the external RIP apparatus **500** as a comparison image in the defect inspection **233** in the first exemplary embodiment.

<Defect Inspection>

[0129] Defect inspection executed by the CPU **223** in step S**2010** in a third exemplary embodiment will be described with reference to FIG. **15**.

[0130] In step S**8001**, the CPU **223** reads the second and third image data from the storage device **222**.

[0131] In step S**8002**, the CPU **223** performs positional adjustment processing so that a position of the third image data is matched with a position of the second image data. As a method of positional adjustment processing, it is possible to perform positional adjustment by detecting a position of an end portion of paper for the third image data and rotating the third image data using an affine transformation method or the like so that the position of the end portion of paper for the third image data is matched with an assumed end portion of an image in the second image data.

[0132] Since processing in steps S**8003** to S**8007** is identical to the processing in steps S**6003** to S**6007**, a detailed description thereof is omitted.

[0133] As described above, the configuration of performing the RIP inspection and the defect inspection according to the third exemplary embodiment makes it possible to output a printed product without a defect as intended by the user and increase reliability of the print apparatus **100**.

Other Exemplary Embodiments

[0134] The description has been given above while various examples and various exemplary embodiments of the present invention are given, the gist and scope of the present invention are not limited to specific description in the present specification.

[0135] The present invention can also be implemented by supply of a program that implements one or more functions of the above-mentioned exemplary embodiments in a system or an apparatus via a network or a storage medium, and readout of the program and execution of processing of the program by one or more processors in a computer of the system or the apparatus. Furthermore, the present invention can also be implemented by a circuit (for example, an application-specific integrated circuit (ASIC)) that implements one or more functions.

[0136] The present invention is not limited to the above-mentioned exemplary embodiments, and can be changed and modified in various manners without departing from the spirit and range of the present invention. Thus, the claims are attached hereto to publicize the scope of the present invention.

Other Embodiments

[0137] Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.

[0138] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

Claims

1. A quality inspection apparatus capable of communicating with at least an image forming apparatus, the quality inspection apparatus comprising: a reception interface (IF) configured to receive from the image forming apparatus a first image generated by execution of image generation processing on print data by the image forming apparatus; a controller having a processor which executes instructions stored in a memory or having circuitry, the controller being configured to: execute first inspection based on a scanned image acquired by reading of a printed product printed by the image forming apparatus and the first image; and execute second inspection based on a second image generated by execution of image generation processing on the print data by an image processing apparatus and the first image; and a transmission IF configured to transmit information based on an inspection result of the first inspection and an inspection result of the second inspection to the image forming apparatus.

2. The quality inspection apparatus according to claim 1, wherein the controller is configured to compare the scanned image and the first image to inspect print smudges caused in print performed by the image forming apparatus.

3. The quality inspection apparatus according to claim 1, wherein the controller is configured to compare the first image and the second image to inspect a positional shift that is a difference between image data generated by execution of image generation processing by the image forming apparatus and image data generated by execution of image generation processing by the image processing apparatus.

4. The quality inspection apparatus according to claim 1, wherein the controller is configured to compare the first image and the second image to inspect a color difference that is a difference between image data generated by execution of image generation processing by the image forming apparatus and image data generated by execution of image generation processing by the image

processing apparatus.
5. The quality inspection apparatus according to claim 1, further comprising a display, wherein the second inspection is either positional shift inspection or color inspection or both the positional shift inspection and the color inspection, and wherein, on a screen of the display on which the first image is displayed, both a range in which the positional shift inspection is executed and a range in which the color inspection is executed are designated or either the range in which the positional shift inspection is executed or the range in which the color inspection is executed is designated.
6. The quality inspection apparatus according to claim 5, wherein an inspection level is settable in a range in which the second inspection is executed.
7. The quality inspection apparatus according to claim 6, wherein the inspection level is a threshold for calculating an inspection result by comparison with a difference calculated from the first image and the second image.
8. The quality inspection apparatus according to claim 1, further comprising a display, wherein the controller is configured to superimpose both a positional shift and a color difference that are detected by the second inspection or either the positional shift or the color difference that is detected by the second inspection on a screen of the display on which the first image is displayed.
9. The quality inspection apparatus according to claim 1, further comprising a display, wherein the controller is configured to superimpose both a positional shift and a color difference that are detected by the second inspection or either the positional shift or the color difference that is detected by the second inspection on a screen of the display on which the second image is displayed.
10. The quality inspection apparatus according to claim 1, wherein the quality inspection apparatus is communicably connected to the image processing apparatus, and wherein the quality inspection apparatus further comprises another reception IF configured to receive the second image generated by execution of image generation processing on the print data by the image processing apparatus.
11. The quality inspection apparatus according to claim 1, wherein the information based on the inspection result of the first inspection and the inspection result of the second inspection is information to control a paper discharge destination of the printed product and information to control an operation of the image forming apparatus.
12. The quality inspection apparatus according to claim 11, wherein the information is information to stop print performed by the image forming apparatus in a case where the inspection result of the second inspection is a fail.
13. An inspection system in which at least an image forming apparatus, an image processing apparatus, and a quality inspection apparatus are communicably connected with one another, the inspection system comprising: a registration unit configured to register a first image generated by execution of image generation processing on print data by the image forming apparatus; a print unit configured to print the first image on a recording sheet; a reading unit configured to read a printed product printed by the print unit and generate a scanned image; a first inspection unit configured to compare the scanned image acquired by the reading unit and the first image registered by the registration unit; a second inspection unit configured to compare a second image generated by execution of image generation processing on the print data by the image processing apparatus and the first image; and a control unit configured to control a paper discharge destination of the printed product based on a result of the comparison made by the first inspection unit and a result of the comparison made by the second inspection unit.
14. The inspection system according to claim 13, wherein the first inspection unit is configured to compare the scanned image and the first image to inspect print smudges caused in print performed by the print unit.
15. The inspection system according to claim 13, wherein the second inspection unit is configured to compare the first image and the second image to inspect a positional shift and a color difference between image data generated by execution of image generation processing by the image forming

apparatus and image data generated by execution of image generation processing by the image processing apparatus.

16. The inspection system according to claim 13, wherein, in a case where a result of inspection performed by the first inspection unit is OK, the control unit is configured to differentiate a paper discharge destination of the printed product depending on a result of inspection performed by the second inspection unit.

17. The inspection system according to claim 13, wherein, in a case where both a positional shift and a color difference are detected or either the positional shift or the color difference is detected as a result of the comparison made by the second inspection unit, the control unit is configured to perform control to stop print performed by the image forming apparatus.

18. The inspection system according to claim 13, wherein, in a case where a result of inspection performed by the first inspection unit is a fail and a result of inspection performed by the second inspection unit is OK, the control unit is configured to control the image forming apparatus to perform re-print.
